

Sri Harsha Sukhamanchi

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PROFESSIONAL SUMMARY

- **Software Engineer with around 4 years of experience** designing, developing, modernizing, and supporting **enterprise-scale backend systems, distributed applications, data platforms, and GenAI/LLM-powered solutions** across content publishing, AI, retail, and enterprise data domains.
- Strong expertise in **Java 17/21, Spring Boot, Spring WebFlux, REST APIs, microservices, Kafka, Kafka Streams, MongoDB, Apache Solr, Redis, and event-driven architectures**, with experience building scalable, resilient, high-performance backend services and real-time data processing solutions.
- Hands-on experience building **GenAI/LLM-powered applications and data services**, including end-to-end inference pipelines for **content review, policy violation detection, fix recommendations, intent extraction, spellcheck, image accessibility validation, AI-assisted workflows, and streaming AI responses**, with experience integrating AI engines into distributed backend systems.
- Experienced in **cloud and data engineering across AWS and Azure**, including **Amazon S3, STS, IAM, IAM Roles Anywhere, KMS, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure SQL, Spark, SQL, Python, and ETL pipelines**, with hands-on experience in data modeling, validation, indexing, query optimization, and large-scale data processing.
- Proven expertise in **platform modernization, performance engineering, DevOps, CI/CD, and production support**, including **Java/Jakarta EE migrations, Maven dependency management, vulnerability remediation, secure cloud deployments, RIO, Ansible, release cutovers, UAT deployments, observability, root-cause analysis, resilience engineering, and performance optimization**.
- Strong experience designing and optimizing **high-throughput and resilient systems**, including non-blocking reactive services, circuit breakers, automated retries, exponential backoff, rate-limit handling, database optimization, real-time streaming pipelines, and performance improvements validated through production-oriented metrics.
- Experienced in **Agile/Scrum and cross-functional engineering environments**, collaborating with **product, engineering, QA, DevOps, cloud, security, and infrastructure teams** to troubleshoot complex distributed-system issues, deliver production-ready solutions, and support enterprise releases.

EDUCATION

University of Delaware

Master of Science Data Science, CGPA – 3.8/4.0

Delaware, DE

Feb 2023 – Feb 2025

GMR Institute of Technology

Bachelor of Technology, Major in Computer Science, CGPA – 3.3/4.0

Srikakulam, India

Aug 2017 – Aug 2021

SKILLS

Programming Languages: Java, Python, SQL, R

Frameworks & Libraries: Spring Boot, Spring WebFlux, Resilience4j, JUnit 5, Mockito, TensorFlow, PyTorch, Keras, Scikit-Learn, Pandas, NumPy

Generative AI & Machine Learning: LLMs, RAG, Prompt Engineering, Google Gemini, Predictive Modeling, Classification, Regression, Clustering, Neural Networks, Time Series Analysis, Statistical Analysis

Data Engineering: PySpark, Apache Kafka, MongoDB, MySQL

Cloud Platforms: AWS (S3, IAM, KMS, STS, IAM Roles Anywhere), Azure (Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure Data Lake, Azure SQL)

DevOps & Deployment: Docker, Maven, Git, GitHub Enterprise, Ansible, CI/CD Pipelines, Multi-Environment Deployments (IT/UAT/PROD)

Monitoring & Observability: Splunk, Prometheus, Shield

Architecture & Development: Microservices, REST APIs, Event-Driven Architecture, System Design, Low-Level Design (LLD)

Tools: Tableau, Teradata SQL Assistant, Visual Studio Code, RStudio

CERTIFICATIONS

- Microsoft: AZ-900, Microsoft: DP-900, Microsoft: DP-203

WORK EXPERIENCE

Java / GenAI Developer | Infosys (Client: Apple Inc.)

Nov 2025 – Present

- Led backend development and modernization of **enterprise content publishing microservices** using **Java 21, Spring Boot, MongoDB, Kafka, Apache Solr, Redis, and REST APIs**, supporting scalable content authoring, publishing, translation, workflow, and asset-management capabilities.
- Designed and enhanced **event-driven publishing workflows** using Kafka and Kafka Streams, including sequential locale processing and phase-based orchestration to reduce redundant processing and improve workflow efficiency.
- Executed platform modernization across **12+ production services**, including **Java 17→21 migration, Jakarta EE migration, JUnit modernization, and Maven dependency optimization**, while resolving compatibility, build, and test issues.
- Developed and enhanced **Apache Solr-based text search and filtering** capabilities, supporting content discovery and search workflows within the enterprise content platform.
- Engineered secure **AWS S3-based asset delivery** using IAM Roles Anywhere, KMS, feature flags, and controlled service-to-service access, supporting migration from legacy asset-delivery mechanisms.
- Strengthened application security and reliability by remediating **critical CVEs, dependency vulnerabilities, hardcoded secrets, and transitive dependency issues** across multi-module applications.
- Maintained **CI/CD and deployment automation** using R10, Fortress, and Ansible, supporting multi-environment releases, UAT deployments, release cutovers, and rollback capabilities.
- Performed **release cutover activities across multiple services**, including POM version updates, release-branch preparation, dependency alignment, build validation, artifact coordination, and promotion of compatible service versions to UAT.
- Engineered performance optimizations for secure digital asset delivery, reducing processing time from **14.13 seconds to 3.73 seconds (~74% reduction)** for 150 assets through batch-size tuning, request-level instrumentation, and root-cause analysis of the signing workflow.
- Designed and validated a **MongoDB indexing strategy** for workflow-step search, reducing documents examined from **142,252 to 1,857 (~98.7% reduction)** and query execution time from **286 ms to 7 ms (~97.6% reduction)** using explain() analysis.
- Investigated and resolved complex **production and UAT defects** across distributed services involving authentication, authorization, workflow processing, publishing, translation, auditing, versioning, and asset management through root-cause analysis and cross-service troubleshooting.
- Collaborated with **cloud, security, DevOps, QA, and infrastructure teams** to resolve IAM, KMS, access-control, deployment, and service-integration issues and ensure successful enterprise releases.

Data Engineer | Nexus AI Solutions

Apr 2025 – Nov 2025

- **Developed the real-time data streaming pipeline for an enterprise AI chatbot using Java and Spring Boot**, engineering a core conversation API to efficiently process user prompts and ingest structured interaction history into a **cloud data tier**.
- **Integrated the Endor AI engine into backend data services using Spring WebFlux**, enabling high-throughput, non-blocking delivery of streaming AI text responses across **distributed cloud infrastructure**.
- **Eliminated critical production HTTP 429 (Too Many Requests) pipeline failures** by implementing a robust resilience layer using **Resilience4j** for circuit breaking and automated retries.
- **Optimized pipeline traffic and mitigated retry storms** against rate-limited AI platforms by configuring exponential backoff with a 50% randomized jitter factor to stagger concurrent database worker threads.
- **Managed the enterprise cloud data layer using Spring Data MongoDB**, utilizing custom repositories and execution templates to run efficient, atomic partial-document updates that minimized database write latency.
- **Designed and deployed a comprehensive Splunk observability dashboard for the cloud data platform**, tracking real-time pipeline performance, system queue sizes, and thread pool saturation metrics.
- **Maintained automated CI/CD pipelines and managed application deployments** across multi-module Java systems onto enterprise **Virtual Machines (VMs) and private cloud environments**.

- **Secured cloud infrastructure deployments using Fortress and Ansible**, automating server configurations and managing environment variables without exposing hardcoded system credentials.
- **Partnered with Quality Engineering (QE) and cloud platform teams** to structurally increase API rate limits and validate end-to-end data processing workflows.
- **Collaborated across backend data engineering teams** to provide technical troubleshooting support, unblock multi-module application defects, and perform routine code reviews.

Software Engineer - Data | TCS (Client: Meijer)

Aug 2021 - Jan 2023

- Developed complex Azure Data Factory, and Azure Synapse Analytics pipelines for structured and unstructured datasets for Enterprise Data Analytics and Retail Media Network programs.
- Created pipelines and Azure Databricks notebooks for optimal data handling, by maintaining data integrity and accuracy measures in data engineering processes.
- Utilized Data factory and Synapse pipelines to develop and implement robust ETL processes, by successfully migrating data from diverse sources such as Naas location, Teradata, JSON, CSV, etc., to Azure cloud.
- Proficient in developing ETL solutions using Spark, SQL, and Python to extract, transform, and aggregate data from various file formats and sources, enabling the analysis and discovery of customer usage patterns by validation and cleansing of data.
- Composing and optimizing SQL queries in complex, large-scale business environments, achieving exceptional data processing efficiency
- Data Mapping: Creating and maintaining data mapping documents and ETL design specifications to facilitate data integration projects.
- Agile Methodologies: Proficient in Agile methodologies, specifically Scrum and Sprint, applying them to enhance project efficiency and collaboration.
- ETL Design and Development: Led the design and development of complex ETL processes, ensuring data extraction from diverse sources such as Salesforce, CM360, Teradata, Oracle, etc., and subsequent transformation and loading into data warehouses or other target systems.
- Configure CI/CD pipelines for data engineer teams for Databricks notebook deployments using Python & Rest APIs, resulting in a 75% reduction of deployment wait times.
- Designed pipelines and notebooks using optimized queries, resulting in a remarkable 50% reduction in data processing time and a significant 35% decrease in manual error.

Data Engineer Intern | Topnotch

April 2019 - Jun 2019

- Created interactive dashboards with Tableau on Azure, showcasing insights from Azure data solutions.
- Engineered real-time data processing workflows using Azure Blob Storage and Azure Databricks, with data loaded into Azure Synapse Analytics for comprehensive analysis.
- Expertly utilized SQL and Azure SQL Database for data validation, manipulation, and management of data inconsistencies across Azure database.